

The Variational Method in Quantum Mechanics

Before we start looking at any approximation methods in quantum mechanics, we will explore a rather convenient method to “guesstimate” the ground state energy of a quantum system. This method is called the *variational method*, and hopefully that nomenclature will make sense soon. This discussion will also be a refresher of some solutions to Schrödinger’s equation you have already seen in QM-I.



Readings of S&N Chapter 5.4 and/or Shankar Chapter 16.1. Our homework will be based on Shankar’s example.

The Variational Principle

The variational principle states that for any normalized trial wave function ψ_{trial} , the expectation value of the Hamiltonian is always greater than or equal to the ground state energy E_0 of the system:

$$E_{\text{trial}} = \langle \hat{H} \rangle_{\psi_{\text{trial}}} = \frac{\langle \psi_{\text{trial}} | \hat{H} | \psi_{\text{trial}} \rangle}{\langle \psi_{\text{trial}} | \psi_{\text{trial}} \rangle} \geq E_0. \quad (1)$$



Prove this by expanding the trial wavefunction in the energy eigenbasis and using the fact that all energy eigenvalues are greater than or equal to E_0 .

This means that any trial wave function we choose will give us an upper bound on the ground state energy of the system, E_0 . It really does not matter what trial wave function we choose. The result in Eq. (1) holds. However, it does not tell us how close we are to the true ground state energy, which is the question we really care about. Had we picked the exact ground state wave function as our trial wave function, we would have obtained the exact ground state energy (at that point we would already know the answer, which is not the case in practice).

The “variational” adjective of the method comes in now: once we decided on a functional form for our trial wavefunction, it typically has one or more free parameters. For example, if we pick a Gaussian trial wavefunction, then its width σ and center position μ are free parameters. The goal will then be to minimize the expectation value E_{trial} by varying these free parameters. In doing so, we can obtain the best possible estimate for the ground state energy within the chosen functional form of the trial wave functions. Mathematically, if our trial wave function depends on a set of parameters $\{\alpha_i\}$ (for the Gaussian, think $\{\alpha_i\} = \{\mu, \sigma\}$), we can write:

$$E_{\text{trial}}(\{\alpha_i\}) = \frac{\langle \psi_{\text{trial}}(\{\alpha_i\}) | \hat{H} | \psi_{\text{trial}}(\{\alpha_i\}) \rangle}{\langle \psi_{\text{trial}}(\{\alpha_i\}) | \psi_{\text{trial}}(\{\alpha_i\}) \rangle}. \quad (2)$$

By minimizing $E_{\text{trial}}(\{\alpha_i\})$ with respect to the parameters $\{\alpha_i\}$, we can find the best approximation to the ground state energy:

$$E_0 \leq \min_{\{\alpha_i\}} E_{\text{trial}}(\{\alpha_i\}). \quad (3)$$

Hence, the the best solution here corresponds to the lowest energy we can achieve by varying the parameters of our trial wave function.

The art of the variational method is to pick a trial wave function that is as close as possible to the true ground state wave function, while still being simple enough to compute the expectation value of the Hamiltonian. The more motivated by the physics of the problem at hand and mathematically tractable the trial wavefunction is, the more effective the method will be.



Before looking at application, we note that in practice, the upper bound provided by E_{trial} is best calculated in a specific basis. In position space, for example, we can insert the completeness relation to find:

$$E_{\text{trial}} = \frac{\langle \psi_{\text{trial}} | \hat{H} (\int d^3\mathbf{x} |\mathbf{x}\rangle \langle \mathbf{x}|) | \psi_{\text{trial}} \rangle}{\langle \psi_{\text{trial}} | (\int d^3\mathbf{x}' |\mathbf{x}'\rangle \langle \mathbf{x}'|) | \psi_{\text{trial}} \rangle} = \frac{\int d^3\mathbf{x} \psi_{\text{trial}}^*(\mathbf{x}) \hat{H} \psi_{\text{trial}}(\mathbf{x})}{\int d^3\mathbf{x}' |\psi_{\text{trial}}(\mathbf{x}')|^2}. \quad (4)$$

What basis we choose to work with is up to us and the best choice will depend on the Hamiltonian of the system and the basis in which it is the simplest to write down.

The Hydrogen Atom

As an example, let us use the variational method to estimate the ground state energy of the hydrogen atom. You already know the answer:

$$E_0 = -\frac{\alpha}{2a_B} \simeq -13.6 \text{ eV} \quad (5)$$

where $a_B = \hbar^2/(m\alpha)$ is the Bohr radius and $\alpha = e^2/(4\pi\epsilon_0)$ the fine structure constant. Obviously this is negative because the electron is bound to the proton. In what follows, we will illustrate the variational method by “rediscovering” this result and then by crudely approximating it.

The Hamiltonian of the hydrogen atom is given by

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 - \frac{\alpha}{r} \quad (6)$$

where m is the mass of the electron, e is the elementary charge, and r is the distance from the proton (assumed to be at the origin). We already know the solutions to this problem. They are of the form:

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell}^m(\theta, \phi) \quad (7)$$

where $R_{n\ell}(r)$ are the radial wave functions and $Y_{\ell}^m(\theta, \phi)$ are the spherical harmonics. The ground state wave function is given by

$$\psi_{100}(r, \theta, \phi) = \frac{1}{\sqrt{\pi a_B^3}} e^{-r/a_B} \quad (8)$$

with energy E_0 given by Eq. (5).

The lucky guess

Suppose now that by some miracle, we guessed the exact functional form for the ground state wave function as our trial wave function:

$$\psi_{\text{trial}}(r, \theta, \phi) = C e^{-r/a}, \quad (9)$$

where C is some arbitrary normalization constant and a is some parameter we are allowed to vary. We know that the right answer is $a = a_B$, but let us pretend we do not know that yet. Then, we can compute the expectation value of the Hamiltonian using Eq. (4):

$$E_{\text{trial}}(a) = \frac{\int d^3\mathbf{x} \psi_{\text{trial}}^*(\mathbf{x}) \hat{H} \psi_{\text{trial}}(\mathbf{x})}{\int d^3\mathbf{x}' |\psi_{\text{trial}}(\mathbf{x}')|^2}, \quad (10)$$

and then minimize $E_{\text{trial}}(a)$ with respect to a to find the value of a that best approximates the ground state energy (in this case, it will be exactly the ground state energy).

We will need the Hamiltonian acting on the trial wave function:

$$\begin{aligned}
\psi_{\text{trial}}^*(\mathbf{x}) \hat{H} \psi_{\text{trial}}(r) &= C^2 e^{-r/a} \left(-\frac{\hbar^2}{2m} \nabla^2 - \frac{\alpha}{r} \right) e^{-r/a} \\
&= C^2 e^{-r/a} \left(-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} e^{-r/a} \right) - \frac{\alpha}{r} e^{-r/a} \right) \\
&= C^2 e^{-2r/a} \left(\frac{\hbar^2}{2ma^2} (2ar - r^2) - \alpha r \right)
\end{aligned} \tag{11}$$

which will be integrated over all space for the numerator if Eq. (10):

$$4\pi C^2 \int_0^\infty dr r^2 e^{-2r/a} \left(\frac{\hbar^2}{2ma^2} (2ar - r^2) - \alpha r \right) = 4\pi C^2 \left(\frac{a\hbar^2}{8m} - \frac{a^2\alpha}{4} \right). \tag{12}$$

and the denominator normalization integral is just $C^2 \int d^3\mathbf{x}' e^{-2r/a} = C^2 4\pi \int_0^\infty dr r^2 e^{-2r/a} = C^2 \pi a^3$, finally giving the simple expression

$$E_{\text{trial}}(a) = \frac{\hbar^2}{2ma^2} - \frac{\alpha}{a}. \tag{13}$$

This energy expectation value makes physical sense. It displays a competition between the kinetic energy term ($1/a^2$) and the potential energy term ($-1/a$). If our trial wavefunction is too localized (small a), the kinetic energy will dominate and increase the total energy. On the other hand, if our trial wavefunction is too spread out (large a), the kinetic energy is small, but the potential energy is not as negative as it could be.

Let us now minimize this expression with respect to a :

$$\frac{dE_{\text{trial}}}{da} = -\frac{\hbar^2}{ma^3} + \frac{\alpha}{a^2} = 0 \implies \boxed{a = \frac{\hbar^2}{m\alpha} = a_B}. \tag{14}$$

And so we recover the Bohr radius as the optimal variational parameter. Plugging this value of a back into $E_{\text{trial}}(a)$, we find

$$\boxed{E_{\text{trial}}(a_B) = -\frac{\alpha}{2a_B} = E_0}, \tag{15}$$

which is exactly the ground state energy of the hydrogen atom from Eq. (5).

An “asymptotically lucky” guess

Now, a more realistic scenario is that we do not know the exact functional form of the ground state wave function. Let us pretend we do not know this solution and try to estimate the ground state energy using a “bad” guess. By physical intuition, we can expect that the ground state wavefunction ought to 1) be spherically symmetric, 2) decrease with distance r , and 3) not be singular at the origin. These criteria are calling for our known exponential solution, but we are stubborn and ignore it. Instead, we try something different, namely an inverse power law:

$$\psi_{\text{trial}}(r) = \frac{C}{(r_0 + r)^n}, \tag{16}$$

where C is a normalization constant, n is the power law exponent, and r_0 regularizes the wavefunction so it does not explode at $r = 0$. We will treat n and r_0 as variational parameters to be optimized. Before proceeding, we got to make sure that this wave function is normalizable, as otherwise, our denominator in Eq. (4) would be ill-defined. This is easy to check:

$$\int |\psi_{\text{trial}}(r)|^2 d^3r = \int d\Omega dr r^2 |\psi_{\text{trial}}(r)|^2 = 4\pi |C|^2 \int_0^\infty dr \frac{r^2}{(r_0 + r)^{2n}} \stackrel{!}{=} 1, \tag{17}$$

which shows that for large r , this integral converges only if $2-2n < -1$, or equivalently, $n > 3/2$. In passing, we note that

$$C = \sqrt{\frac{(2n-3)(n-1)(2n-1)}{4\pi r_0^{3-2n}}}. \quad (18)$$

Keeping the requirement of $n > 3/2$ in mind, we can now apply the variational method to compute the expectation value of the Hamiltonian on our trial wave function. Using Eq. (4), we have

$$E_{\text{trial}} = \frac{\int d^3\mathbf{x} \frac{C}{(r_0+r)^n} \left(-\frac{\hbar^2}{2m} \nabla^2 - \frac{\alpha}{r}\right) \frac{C}{(r_0+r)^n}}{\int d^3\mathbf{x}' \left|\frac{C}{(r_0+r)^n}\right|^2}, \quad (19)$$

and `mathematica` will spare us from the mess and already give you the numerator:

$$\begin{aligned} E_{\text{trial}} &= \left(\frac{\hbar^2}{2m} \frac{n r_0^{1-2n}}{(2n-1)(1+2n)} - \alpha \frac{r_0^{2-2n}}{2(n-1)(2n-1)} \right) \left(\frac{r_0^{3-2n}}{(n-1)(2n-3)(2n-1)} \right)^{-1} \\ &= \frac{\hbar^2}{2m} \frac{n(2n-3)(n-1)}{(1+2n)} \frac{1}{r_0^2} - \alpha \frac{(2n-3)}{2r_0}. \end{aligned} \quad (20)$$

Again we find the competition between kinetic and potential energy. We would this to be as negative as possible, so we want to minimize $E_{\text{trial}}(r_0, n)$ with respect to both r_0 and n . Let us keep things simple and just minimize $E_{\text{trial}}(r_0, n)$ with respect to r_0 and see what we get after by playing with values of $n > 3/2$. We will need:

$$\frac{\partial E_{\text{trial}}}{\partial r_0} = -\frac{\hbar^2}{m} \frac{n(2n-3)(n-1)}{(1+2n)} \frac{1}{r_0^3} + \alpha \frac{(2n-3)}{2r_0^2} = 0. \quad (21)$$

which gives us the optimal r_0 :

$$\boxed{r_0^{\text{optimal}} = \frac{\hbar^2}{m\alpha} \frac{n(n-1)}{(\frac{1}{2}+n)} = a_B \frac{n(n-1)}{(\frac{1}{2}+n)}}. \quad (22)$$

Using the ground state energy definition from Eq. (5), we can write the trial energy at this optimal r_0 as

$$\boxed{E_{\text{optimal}} = -\frac{m\alpha^2}{8\hbar^2} \frac{(2n-3)(1+2n)}{n(n-1)} = E_0 \left(\frac{(n-\frac{3}{2})(n+\frac{1}{2})}{n(n-1)} \right) \xrightarrow{n \rightarrow \infty} E_0}. \quad (23)$$

Clearly, as shown in Fig. 1, the best choice of n is just the $n \rightarrow \infty$ limit, which gives us $E_{\text{trial}} \rightarrow E_0$. On the other hand, for n close to the normalizability limit of $3/2$, the variational estimate is quite poor, even predicting unbound states with $E_{\text{trial}} \geq 0$ for $n \lesssim 3/2$. A peculiarity here is that as $n \rightarrow \infty$, the optimal radius goes to infinity.

There is an easy way to understand why $n \rightarrow \infty$ is optimal. First, let us rewrite our fully normalized trial wave function:

$$\psi_{\text{trial}}(t) = \sqrt{\frac{(2n-3)(n-1)(2n-1)}{4\pi r_0^3}} \left(1 + \frac{r}{r_0}\right)^{-n}. \quad (24)$$

This is the same as Eq. (16) but with the normalization constant C explicitly written out. Now, we plug in our optimal r_0 . The result is kind of a mess, so let us do so in the regime where $n \gg 1$, where $r_0^{\text{optimal}} \approx a_B n$. In that case,

$$\psi_{\text{trial}}(r) \xrightarrow{n \gg 1} \frac{1}{\sqrt{\pi a_B^3}} \left(1 + \frac{r}{a_B n}\right)^{-n}. \quad (25)$$

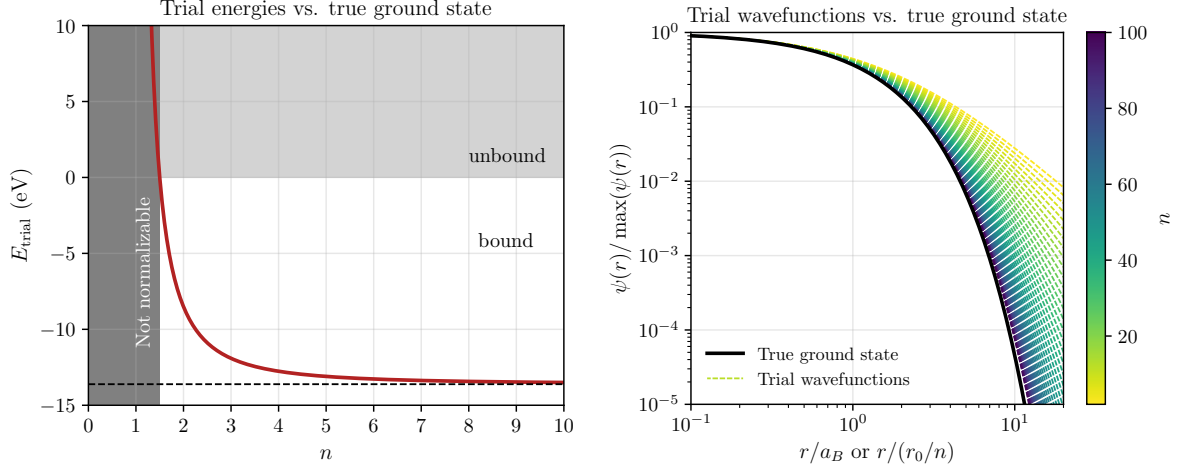


Figure 1: Left panel) Estimate of the hydrogen ground state energy as a function of the power law exponent n of our trial wave function. The horizontal dashed line indicates the true ground state energy E_0 . Shaded regions are unphysical. Right panel) Comparison of the trial wave functions for different n values (colored dashed lines) against the true hydrogen ground state wave function (black solid line).

This is looking more and more like Eq. (8). But it gets better. Using the limit definition of the exponential function, we have

$$\lim_{n \rightarrow \infty} \left(1 + \frac{r}{a_B n} \right)^{-n} = e^{-r/a_B}, \quad (26)$$

and so we see that in the $n \rightarrow \infty$ limit, our trial wave function converges to the true ground state wave function of the hydrogen atom! This is exactly what we are seeing in the right panel of Fig. 1, where the trial wave function approaches the true ground state wave function as n increases (but only after interpreting the new Bohr radius as r_0/n). This explains why the variational energy estimate also converges to the true ground state energy in this limit.



Note that the variational method is better at estimating ground state energies than wave functions. To see this, imagine that the trial wave function is small shift ϵ away from the true ground state, $|\psi_{\text{trial}}\rangle = |\psi_0\rangle + \epsilon |\delta\psi\rangle$. This shift is of $\mathcal{O}(\epsilon)$ in the wave function. However, the error induced on the energy estimate is of $\mathcal{O}(\epsilon^2)$:

$$\delta E \sim E_{\text{trial}} - E_0 \sim \langle \psi_{\text{trial}} | \hat{H} | \psi_{\text{trial}} \rangle - \langle \psi_0 | \hat{H} | \psi_0 \rangle \sim \epsilon^2 \langle \delta\psi | \hat{H} | \delta\psi \rangle. \quad (27)$$

This explains why even crude trial wave functions can yield good energy estimates.